

PROJECT DEVELOPMENT PHASE: MODEL PERFORMANCE TEST

SWTID-2026-8437

Machine Learning Template Alignment – Auto Ticket Classification / Flow Design

DATE	TEAM ID	PROJECT NAME	MAXIMUM MARKS
1.10.2026	SWTID-2026-8437	Machine Learning	10 Marks

Important Scope Note: The Machine Learning Performance Test template evaluates Regression, Classification, Hyperparameter Tuning, and Validation Methods. Because the core project requirement implements a deterministic script-controlled Access Control List (ACL) in Flow Design rather than a probabilistic ML training model, standard ML metrics (MAE, RMSE, Confusion Matrix) are designated as *Not Applicable (N/A)*. Authorization behavior is evaluated via rule-based security test verification.

1. MODEL PERFORMANCE TESTING PARAMETERS

S.No	Parameter	Values & Technical Description	Screenshot / Evidence
1	Metrics – Regression Model	Not Applicable (N/A) MAE: N/A MSE: N/A RMSE: N/A R² Score: N/A <i>Reason:</i> The solution does not predict continuous numerical outputs.	[N/A – No Regression Model]
2	Metrics – Classification Model	Rule-Based Security Decision Matrix • EEE user + Branch = EEE → ALLOW • Non-EEE user + Branch = EEE → DENY • Administrator → ALLOW (Full Access)	[Insert ACL Test Results]
3	Confusion Matrix	Binary Authorization Outcome Mapping Tracks actual vs. expected access decisions (Allowed/Denied) across defined security test cases upon execution.	[Insert Completed Matrix]
4	Accuracy Score	Functional Authorization Accuracy Calculated post-execution: Accuracy = (Correct Authorization Decisions ÷ Executed Cases) × 100	[Insert Test Evidence]
5	Classification Report	Not Applicable as ML Report Precision, Recall, and F1-Score are recorded only if authorization rules are explicitly evaluated as a binary classification matrix.	[N/A or Insert Actual Report]
6	Hyperparameter Tuning	Not Applicable (N/A) No hyperparameter search (e.g., GridSearch/RandomSearch) or learning rate tuning is required for script-controlled ACL rules.	[N/A]
7	Validation Method	Controlled Authorization Scenario Testing Validates authorization across EEE users, non-EEE users, Administrators, and edge/empty branch field conditions.	[Insert UAT / Test Screenshot]

2. PROJECT-SPECIFIC VALIDATION RESULTS

Execution matrix for machine learning model performance evaluation under ticket reference **SWTID-2026-8437**:

Case ID	User Type	Record Branch	Expected Decision	Actual Result	Status
ML-01	EEE User	EEE	ALLOW	To be filled	☐
ML-02	Non-EEE User	EEE	DENY	To be filled	☐
ML-03	Administrator	EEE	ALLOW	To be filled	☐
ML-04	Administrator	Other Branch	ALLOW	To be filled	☐
ML-05	EEE / Non-EEE	Empty / Unexpected	Follow Secure Defined Rule	To be filled	☐

3. AUTHORIZATION VALIDATION ACCURACY FORMULA

Calculation Formula:

Authorization Validation Accuracy (%) = (Number of Correct Decisions ÷ Total Executed Cases) × 100

Reference Note: If 5 valid test cases are executed and all 5 match expected decisions, accuracy is 100%. Final scores must be populated from actual test execution logs.

4. EVIDENCE CHECKLIST

- ☐ ACL configuration screenshot displaying protected operations and underlying condition script
- ☐ EEE user successfully viewing an EEE branch record
- ☐ Non-EEE user denied / restricted from viewing an EEE branch record
- ☐ Administrator user successfully accessing protected EEE records
- ☐ Administrator user successfully accessing other-branch records
- ☐ Completed execution evidence log used to compute final authorization validation accuracy

5. FINAL EVALUATION SIGN-OFF

Evaluation & Grading Sign-Off (Max Marks: 10):

ASSIGNED MARKS / 10

EVALUATOR SIGNATURE

REVIEW DATE